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Title: Jaundice in Children Investigation & Management

Jaundice in children is a common condition and needs to be evaluated carefully to identify the transient self-limited infective liver disease from the more serious conditions.

Guidelines of investigating children with conjugated Hyperbilirubinemia

Children presenting with conjugated hyperbilirubinemia should always be investigated. Investigations are aimed to diagnose the etiology of liver disease and its complications. Based on history and physical examination the following investigations should be considered as initial step:

1. CBC
2. Blood glucose
3. Renal function tests
4. Liver functions tests: ALT, AST, GGT, alkaline phosphatase, albumin and globulin gap
5. Direct and total bilirubin
6. Calcium profile
7. Coagulation profile and INR
8. Ammonia, lactate and venous blood gases.
9. Blood group
10. Fasting ultrasound abdomen (Always with IV 10% dextrose + 0.45% Nacl if deranged coagulation).
11. If infective etiology is considered, include blood culture hepatitis Bs Ag, C antibodies.

Neonatal Jaundice

Neonatal Jaundice or jaundice with in the first year of life is a special category of diseases. It should always be considered as a matter of urgency (needs to be assessed sooner than later during working hours).

In all neonatal jaundice always ask about the color of stools and provide containers for stool samples to be seen by the gastroenterology/surgical team if required.

The following investigations are usually needed in addition to the previous ones. These investigations are mainly based on clinical examination and are only requested once the patient has been seen by the consultant gastroenterologist.

3. Urine organic acids or urinary succinyl acetone (if serum amino acids and AFP are suggestive of a pathology)
4. Serum Alpha one antitrypsin level
5. Alpha fetoprotein
6. Hepatitis B surface antigen (HBsAg)
7. PCR for CMV/EBV and relevant TORCH organisms
8. Thyroid function test
9. Iron profile
10. Sweat test

Normal coagulation values for children

Screening test	Preterm 30-36 wks Within first 72 hrs	Term Within first 72 hrs	Older than 1 year/adults
Prothrombin time sec	13-23	13-17	1-16
Activated partial thromboplastin time sec	35-100	35-70	35-45

Children beyond the first year of life

In older children (beyond first year of life), investigations are totally based on clinical indications and child must have been seen by consultant gastroenterology before getting these tests. These might include:

1. Serum amino acids
2. Ammonia and lactate
3. Serum ceruloplasmin
4. Serum copper levels
5. 24 hr- Urinary copper pre and post D-Penicillamine challenge
6. ANA
7. Anti smooth muscle antibodies and anti liver kindly microsomal antibodies
8. Anti mitochondrial antibodies
9. Immunoglobulin levels
10. Direct Coombs Test-polyvalent
11. Hepatitis B profile
12. Hepatitis C antibodies
13. EBV and CMV serology
14. AFP
15. Alpha-1-antitrypsin

Indications for immediate admission of children with liver disease:

Recent onset hyperbilirubinemia with:

1. Deranged coagulation
2. High GGT with pale stools
3. Bruising
4. Hematemesis or Malena
5. Hypoglycemia
6. Ascites (compromising breathing/discomfort)
7. Encephalopathy
8. Septicemia

- Children with chronic liver disease and known to have coagulopathy need not be admitted for deranged coagulation only. Consultant on-call needs to be notified if child is sick or is for admission.
- Babies with neonatal jaundice and no coagulopathy, encephalopathy or other complications need to be admitted for works up during week days only
- Encephalopathy presents early with lethargy or irritability.

Table 7.6 Clinical stages of hepatic encephalopathy.

Stage	Asterixis	EEG changes	Clinical manifestations
I (prodrome)	Slight	Minimal	Mild intellectual impairment, disturbed sleep–wake cycle
II (impending)	Easily elicited	Usually generalized slowing of rhythm	Drowsiness, confusion, coma, inappropriate behavior, disorientation, mood swings
III (stupor)	Present if patient cooperative	Grossly abnormal slowing	Drowsy, unresponsive to verbal commands, markedly confused, delirious, hyperreflexia, (+) Babinski sign
IV (coma)	Usually absent	Appearance of delta waves, decreased amplitudes	Unconscious, decerebrate or decorticate response to pain present (IVA) or absent (IVB)

Adapted from refs. 4, 35.

(Kelly 2008)

- For all new patients with liver disease and admitted on emergency basis, the gastroenterology on call needs to be informed.

Important issues of management

1. Always inform the gastroenterologist in-charge if a child has been admitted from the emergency room with complicated liver disease.
2. Please try not to use ER as a walk-in liver disease by investigating non-urgent liver disease patients as explained earlier. All liver patients (except the above category) should have a referral online from their respective hospitals. All referrals are viewed by consultant gastroenterologist and dealt with on appropriate basis.
3. The emergency department is not a liver clinic and please do not ask for the gastroenterologist to come to the A/E for counseling on new liver disease.
4. In any patient with liver disease avoid keeping NPO. If you need that always put them on at least 10% dextrose containing fluids for infants. Always monitor blood glucose.
5. Please don't administer Fresh Frozen Plasma or blood before collecting the appropriate samples for investigation as the above products interfere with results.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Text book of Neonatology	Roberton.	1992	
Diseases of the Liver and Biliary System in Children	Kelly	2008	179

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